

Minyoung Choe

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EDUCATION

Korea Advanced Institute of Science and Technology (KAIST)

M.S. & Ph.D. in Artificial Intelligence

Advisor: Prof. Kijung Shin

Seoul, South Korea

Mar 2021 – Present (Expected 2027.02)

Sogang University

B.S. in Computer Science and Engineering

GPA: 4.14 / 4.3 | Graduated Summa Cum Laude — Ranked 1st / 72

Seoul, South Korea

Mar 2017 – Feb 2021

RESEARCH SUMMARY

- **Research Direction:** Graph/hypergraph mining and learning as a foundation for knowledge graph construction, structured agent memory, and automated ML workflows.
- **Structured Modeling Foundation:** Developed methods for analyzing, generating, and learning from graph/hypergraph data that capture group-level interactions beyond pairwise relations, with multiple publications at top data mining and AI venues including KDD, WWW, ICDM, IJCAI, and TKDD.
- **Knowledge Graphs and LLM Agents:** Constructed graph/hypergraph workflow memory from code, Kaggle notebooks, and GitHub repositories to capture reusable knowledge such as entities, relations, workflows, dependencies, and execution context.
- **KG-Augmented Agentic AutoML:** Designed retriever-augmented AutoML agents that use structured memory for retrieval, technique selection, workflow recombination, planning, coding, debugging, and feedback-driven improvement.
- **Core Strength:** Problem formulation, structured construction, and persistent execution — demonstrated through first-author hypergraph research requiring new task definitions, data/graph construction, model design, and evaluation protocols, with recent extensions toward structured memory and AutoML agent workflows.

RESEARCH & INDUSTRY EXPERIENCE

NEC Laboratories America

Research Intern — Data Science & System Security

Princeton, NJ, USA

Jun – Sep 2025

- **Time-Series Foundation Models with Similar-Case Retrieval:** Used time-series foundation models, including VisionTS and Moirai, to retrieve historical cases with embeddings similar to model predictions, and combined them with a multimodal LLM to interpret forecasting results through analogous past patterns.
- **Kaggle/GitHub-to-KG Autonomous ML Agent:** Designed and implemented an AutoML agent system with Planner–Retriever–Coder–Debugger agents that constructs task-specific knowledge graphs from Kaggle competitions, notebooks, and GitHub repositories. The graph memory captures reusable ML know-how, including preprocessing, feature engineering, modeling, ensembling, dependencies, and execution order, enabling agents to retrieve, recombine, and iteratively search for stronger ML pipeline configurations.

Carnegie Mellon University

Visiting Researcher (IITP-funded CMU AI Concentrated Education Program)

Pittsburgh, PA, USA

Aug 2023 – Feb 2024

- **Compact and Principled Hypergraph Generator:** Collaborated with Prof. Christos Faloutsos on a lightweight, mathematically grounded generative model for real-world hypergraphs, designed to reproduce power-law patterns and higher-order group interactions with very few parameters.
- **WWW 2025 Hypergraph Generation Publication:** Developed research that resulted in a WWW 2025 publication on Kronecker generative models for power-law patterns in real-world hypergraphs.

RESEARCH PROJECTS

Scientific Literature QA and Knowledge Structuring — Industry Collaboration (Samsung)

Jan 2026 – Present

- **Realistic Scientific QA Benchmarking:** Formulated condition-aware scientific QA tasks that extend standard fact-recall QA toward realistic R&D evidence reasoning, where answers depend on conditions, contexts, and evidence across multiple papers.
- **Multi-Agent QA Generation and Verification:** Designed and implemented paper-grounded generation and verification pipelines to improve the reliability of condition-aware QA construction with limited domain-expert annotation, filtering unsupported outputs through grounding, evidence-support, and recoverability checks.

AI-based Weather Forecast Support — Korea Meteorological Administration (KMA)

Jul 2021 – Dec 2024

- **Operational Similar-Weather Retrieval:** Led a KMA project that built a real-time satellite video retrieval system to find historical weather situations analogous to current conditions, supporting forecasters in operational decision-making.
- **Label-free Satellite Video Embedding:** Designed a self-supervised embedding pipeline that compresses high-resolution satellite videos into low-dimensional representations without manual similarity labels, enabling scalable retrieval over large meteorological archives.
- **Prediction-augmented Graph Search:** Built an end-to-end framework combining video compression, future-frame prediction, graph-based nearest-neighbor search, ranking, and visualization; this deployed system resulted in a KDD 2025 Industry Track publication as a co-first author [C8].

PUBLICATIONS

(*** = equal contribution; *underline* = Minyoung Choe; [C] = Conference, [J] = Journal, [W] = Workshop)

[C10] Identifying Group Anchors in Real-World Group Interactions Under Label Scarcity

Fanchen Bu, Geon Lee, Minyoung Choe, and Kijung Shin

ICDM 2025

[C9] Attributed Hypergraph Generation with Realistic Interplay Between Structure and Attributes

Jaewan Chun*, Seokbum Yoon*, Minyoung Choe, Geon Lee, and Kijung Shin

ICDM 2025 🏆 Best Paper Award

[C8] SkySearch: Satellite Video Search at Scale

Minyoung Choe*, Geon Lee*, Changhun Han*, Suji Kim, Woong Hu, Hyebeen Hwang, Geunseok Park, Byeongyeon Kim, Hyesook Lee, Ha-Myung Park, and Kijung Shin

KDD 2025 (Industry Track)

[C7] Kronecker Generative Models for Power-Law Patterns in Real-World Hypergraphs

Minyoung Choe, Jihoon Ko, Taehyung Kwon, Kijung Shin, and Christos Faloutsos

WWW 2025

[W4] Graphlets over Time: A New Lens for Temporal Network Analysis

Deukryeol Yoon, Dongjin Lee, Minyoung Choe, Kijung Shin

ECML/PKDD 2024 Workshop

[W3] GraphEHR: Heterogeneous Graph Neural Network for Electronic Health Records

Minyoung Choe*, Juho Jung*, Kushagra Agarwal*, Nivedhitha Dhanasekaran*

IJCAI 2024 Workshop

[J1] Representative and Back-In-Time Sampling from Real-world Hypergraphs

Minyoung Choe, Jaemin Yoo, Geon Lee, Woonsung Baek, U Kang, Kijung Shin

ACM Transactions on Knowledge Discovery from Data (TKDD), 2024

[W2] Temporal Graph Networks for Graph Anomaly Detection in Financial Networks

Yejin Kim*, Youngbin Lee*, Minyoung Choe, Sungju Oh, Yongjae Lee

AAAI 2024 Workshop

[C6] Classification of Edge-dependent Labels of Nodes in Hypergraphs

Minyoung Choe, Sunwoo Kim, Jaemin Yoo, and Kijung Shin

ACM KDD 2023

[C5] How Transitive Are Real-World Group Interactions? - Measurement and Reproduction

Sunwoo Kim, Fanchen Bu, Minyoung Choe, Jaemin Yoo, and Kijung Shin

ACM KDD 2023

[C4] Reciprocity in Directed Hypergraphs: Measures, Findings, and Generators

Sunwoo Kim, [Minyoung Choe](#), Jaemin Yoo, and Kijung Shin

[IEEE ICDM 2022](#)

[C3] HashNWalk: Hash and Random Walk Based Anomaly Detection in Hyperedge Streams

Geon Lee, [Minyoung Choe](#), and Kijung Shin

[IJCAI 2022](#)

[C2] MiDaS: Representative Sampling from Real-world Hypergraphs

[Minyoung Choe](#), Jaemin Yoo, Geon Lee, Woonsung Baek, U Kang, and Kijung Shin

[WWW 2022](#)

[C1] How Do Hyperedges Overlap in Real-World Hypergraphs? - Patterns, Measures, and Generators

Geon Lee*, [Minyoung Choe*](#), and Kijung Shin

[WWW 2021](#)

[W1] Pretraining Neural Architecture Search Controllers with Locality-based Self-Supervised Learning

Kwanghee Choi*, [Minyoung Choe*](#), Hyelee Lee*

[NeurIPS 2020 Workshop](#)

AWARDS & HONORS

Best Paper Award, IEEE ICDM 2025	2025
Student Scholarship Award, The Web Conference (WWW 2021)	2021
Grand Prize, Open-Source Software Contribution (MSIT, Korea)	Nov 2019
National Science & Technology Scholarship (KOSAF)	2019 – 2021
Dean's Honor List — Top 1%, Sogang University	2018 – 2020
Silver Prize, Programming Competition, Sogang University	Nov 2018

TECHNICAL SKILLS

Research Areas: Graph & Hypergraph Mining and Learning, Scientific Evidence Reasoning, Knowledge Graph/Hypergraph Memory, LLM-based Agent Systems, Retrieval-Augmented AI

ML / DL: PyTorch, PyTorch Geometric (PyG), DGL, Graph Neural Networks, Hypergraph Neural Networks, Self-supervised Learning, Representation Learning, Time-Series Foundation Models

LLM / KG / Retrieval Systems: AutoML Multi-agent Systems (Planner–Retriever–Coder–Debugger Agents), Knowledge Graph Construction, Graph/Hypergraph-Augmented Agents, Scientific Evidence Retrieval

Programming: Python, C/C++, Java

Languages: Korean (native), English (professional)

PROFESSIONAL SERVICE

Workshop Co-organizer — PAKDD 2025 Workshop on Graph Learning with Foundation Models, Sydney, Australia Jun 2025

Program Committee / Reviewer — ACM KDD, The Web Conference (WWW), ACM CIKM, PAKDD

TEACHING EXPERIENCE

KAIST — Teaching Assistant

Data Mining and Search (Spring 2022, 2023, 2024) | Mathematics for AI (Fall 2022)

Seoul, South Korea

Mar 2022 – Jun 2024

Sogang University — Teaching Assistant

Computational Thinking

Seoul, South Korea

Sep – Dec 2018